

Worksheet

Computer Network - Week 6 - Intro to Linux & System Calls

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Activity 1

Answers

1. uname

```
(farrelta@ROG)-[~]  
$ uname  
Linux
```

It is for showing the name of the operating system → Linux

2. df

```
(farrelta@ROG)-[~]  
$ df  
Filesystem      1K-blocks      Used Available Use% Mounted on  
none            3967568         0   3967568   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2  
none            3967568         4   3967564   1% /mnt/wsl  
drivers         473406460 304844012 168562448 65% /usr/lib/wsl/drivers  
/dev/sdd        1055762868 25265452 976793944 3% /  
none            3967568        260   3967308   1% /mnt/wslg  
none            3967568         0   3967568   0% /usr/lib/wsl/lib  
rootfs          3962552        2672   3959880   1% /init  
none            3962552         0   3962552   0% /dev  
none            3967568        856   3966712   1% /run  
none            3967568         0   3967568   0% /run/lock  
none            3967568         0   3967568   0% /run/shm  
none            3967568        2124   3965444   1% /mnt/wslg/versions.txt  
none            3967568        2124   3965444   1% /mnt/wslg/doc  
C:\             473406460 304844012 168562448 65% /mnt/c  
D:\             500105212 423654988 76450224 85% /mnt/d  
F:\             25165820 19218036 5947784 77% /mnt/f  
tmpfs           1024         0      1024    0% /run/credentials/systemd-journald.service  
tmpfs           1024         0      1024    0% /run/credentials/getty@tty1.service  
tmpfs           1024         0      1024    0% /run/credentials/console-getty.service  
tmpfs          793512        120   793392   1% /run/user/1000
```

It shows the disk information of the device, because i use Linux WSL, it shows the windows disk information too

3. hostname

```
(farrelta@ROG)~  
$ hostname  
ROG
```

It shows the host name of the computer → ROG

4. **hostname -i**

```
(farrelta@ROG)~  
$ hostname -i  
127.0.1.1
```

It shows the IP address of the device → 127.0.1.1

Activity 2

Answers

1. Make sure **strace** is installed on your device. If not, you can install it using the command: **sudo apt install strace**

```
(farrelta@ROG)~$ sudo apt install strace
[sudo] password for farrelta:
Installing:
  strace

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 1,473 kB
  Space needed: 3,271 kB / 1,000 GB available

Get:1 http://http.kali.org/kali kali-last-snapshot/main amd64 strace amd64 6.16+ds-2 [1,473 kB]
Fetched 1,473 kB in 1min 32s (16.0 kB/s)
Selecting previously unselected package strace.
(Reading database ... 465846 files and directories currently installed.)
Preparing to unpack .../strace_6.16+ds-2_amd64.deb ...
Unpacking strace (6.16+ds-2) ...
Setting up strace (6.16+ds-2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.3.2) ...
```

I successfully install it

2. Run the command **echo hello** and save its output to a file named "**echo.log**" using **strace**. You can do this using the format:

strace -o your_log_filename command_you_want_to_run

```
(farrelta@ROG)~$ strace -o echo.log echo hello
hello

(farrelta@ROG)~$ ls
Desktop  Documents  Downloads  echo.log  Music  Pictures  Public  Templates  Videos
```

After i run the command, it creates the file "echo.log"

3. Filter the word “**hello**” from the **echo.log** file, then take a screenshot of the result. (You can use the **grep** command to do this.)

```
(farrelta@ROG)-[~]  
$ grep "hello" echo.log  
execve("/usr/bin/echo", ["echo", "hello"], 0x7ffcf3a53d8 /* 33 vars */) = 0  
write(1, "hello\n", 6) = 6
```

Output:

```
execve("/usr/bin/echo", ["echo", "hello"], 0x7ffcf3a53d8 /* 33 vars */) = 0  
write(1, "hello\n", 6) = 6
```

4. Explain what the command you ran in step 3 does. You may refer to the command’s manual page as a reference.

```
(farrelta@ROG)-[~]  
$ grep --help  
Usage: grep [OPTION]... PATTERNS [FILE]...  
Search for PATTERNS in each FILE.  
Example: grep -i 'hello world' menu.h main.c  
PATTERNS can contain multiple patterns separated by newlines.  
  
Pattern selection and interpretation:  
-E, --extended-regexp    PATTERNS are extended regular expressions  
-F, --fixed-strings      PATTERNS are strings  
-G, --basic-regexp       PATTERNS are basic regular expressions  
-P, --perl-regexp        PATTERNS are Perl regular expressions  
-e, --regexp=PATTERNS    use PATTERNS for matching  
-f, --file=FILE          take PATTERNS from FILE  
-i, --ignore-case        ignore case distinctions in patterns and data  
                        --no-ignore-case    do not ignore case distinctions (default)  
-w, --word-regexp        match only whole words  
-x, --line-regexp        match only whole lines  
-z, --null-data          a data line ends in 0 byte, not newline  
  
Miscellaneous:  
-s, --no-messages        suppress error messages  
-v, --invert-match        select non-matching lines  
-V, --version            display version information and exit  
                        --help            display this help text and exit  
  
Output control:  
-m, --max-count=NUM      stop after NUM selected lines
```

It uses to search the pattern in the file